Theorem 15: NP-hardness of the optimal-transport edit distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Theorem16_EditHardness`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper. Theorem 15 (Section 4, p. 12, l. 418; proof in App. G, pp. 50–51, l. 1577–1620): *GSim*$(d_{\mathrm{ed}}^{\mathrm{OT}})$ *is NP-hard, even when restricted to collections of paths.* It is part (1) of Theorem 4 (p. 4, l. 134). The proof is the reduction of [GRW18, Theorem 8] from the strongly NP-hard problem Three-Partition: an instance $(A, a_1, \dots, a_{3m})$ in unary with $\sum_i a_i = mA$ and $A/4 < a_i < A/2$ is mapped to (F_1, F_2, t), where F_1 is the disjoint union of paths on a_i vertices, F_2 the disjoint union of m paths on A vertices, and $t = 4m$. The claim (p. 50, l. 1584–1585) is that $d_{\mathrm{ed}}^{\mathrm{OT}}(F_1, F_2) \le t$ iff the instance is a YES-instance.

Lean. Only the correctness of the reduction is formalised. With $F_1 =$ `pathUnion I.a` and $F_2 =$ `pathUnion fun _ : Fin I.m => I.A`:

- `GsimOT.Hardness.ThreePartition.dOTEdit_le_iff`: if $a_i \ge 2$ for all i, then

$$d_{\mathrm{ed}}^{\mathrm{OT}}(F_1, F_2) \le 4m \iff I \text{ is a YES-instance.}$$

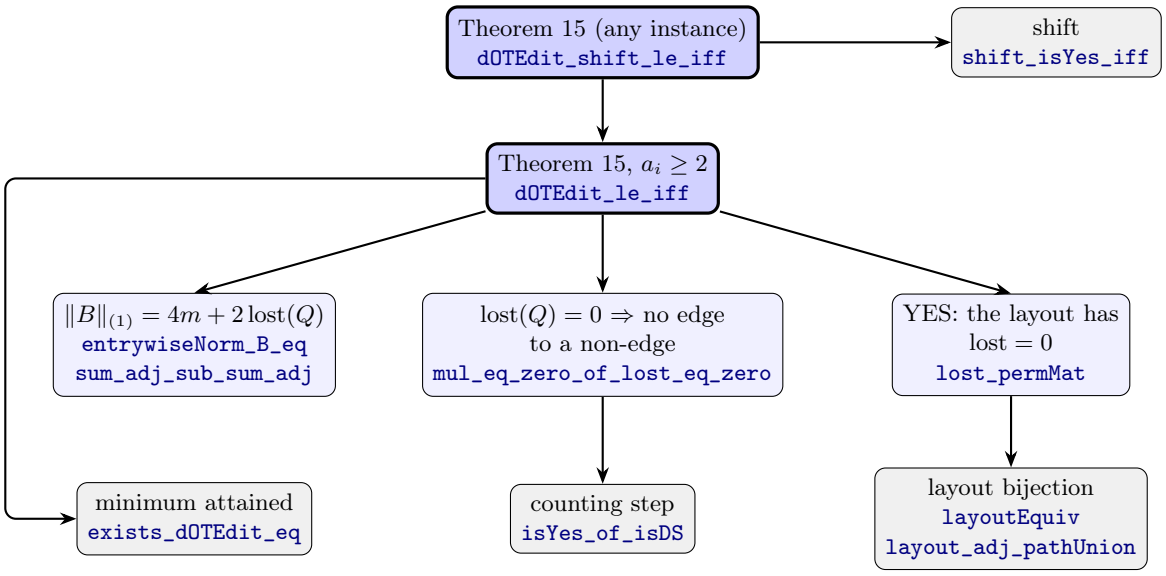
- `GsimOT.Hardness.ThreePartition.dOTEdit_shift_le_iff`: for *every* instance I, the same equivalence for the shifted instance (add 3 to each a_i and 9 to A, l. 1586–1587), whose YES/NO status equals that of I.

Here $d_{\mathrm{ed}}^{\mathrm{OT}}(G, H) = \min_{Q \in \mathrm{DS}(V,W)} \|B(G, H, Q)\|_{(1)}$ is `GsimOT.dOTEdit` (`Common/EditDistance.lean`), the same definition used by Proposition 5 and Theorem 6; a Three-Partition instance is the structure `GsimOT.Hardness.ThreePa` (fields `sum_eq`, `lt_four_mul`: $A < 4a_i$, `two_mul_lt`: $2a_i < A$), and `IsYes` asks for $\sigma\colon [3m] \to [m]$ with classes of size 3 and sum A.

2 Where the proof lives

File	Content
`Theorem16_EditHardness/Theorem16.lean`	$C_{\mathrm{ed}}(Q) = \|B(G,H,Q)\|_{(1)}$, the identity $\|B\|_{(1)} = 4m + 2\,\mathrm{lost}(Q)$, both directions of the reduction, the shifted version.
`Common/ThreePartition.lean`	Three-Partition instances, the layout bijection of a solution (`layoutEquiv`), the counting step `isYes_of_isDS`, the shift.
`Common/PathsAndCycles.lean`	`pathUnion`, edge count `sum_adj_pathUnion`, neighbours (`exists_pathUnion_adj`), constancy along paths (`eq_of_pathUnion_adj`).
`Common/EditDistance.lean`, `Common/MatrixNorms.lean`	`dOTEdit`, `dOTEdit_le`, attainment `exists_dOTEdit_eq`, $C_{\mathrm{ed}}(Q) = \|B(G,H,Q)\|_{(1)}$ (`costEdit_eq_entrywiseNorm_B`).

3 Proof layout



Blue: this folder; grey: `GsimOT/Common`. Nothing is cited.

4 The proof step by step

Step 0: w.l.o.g. $a_i \geq 4$ (l. 1586–1587). The paper adds 3 to every a_i and 9 to A. Lean: `ThreePartition.shift`, `four_le_shift_a`, and `shift_isYes_iff` (each class of a solution has exactly three elements, `card_eq_three_of_sum_eq`, so the class sums shift by 9). The formalised reduction `dOTEdit_le_iff` assumes $a_i \geq 2$, so that every vertex of F_1 has a neighbour (`exists_pathUnion_adj`).

Step 1: passing to blowups (l. 1585–1586). The paper uses $d_{\mathrm{ed}}^{\mathrm{OT}}(F_1, F_2) = d_{\mathrm{ed}}^{\bullet}(F_1, F_2)$, cited from [Gro25, Theorem 3.10], and argues with alignments of blowups. *The formalisation skips this step* and works with $Q \in \mathrm{DS}(V, W)$ directly; see Section 6.

Step 2: YES-instances (l. 1588–1597). A solution σ lays the paths of F_1 out consecutively on the paths of F_2; the bijection ρ maps edges to edges, and only the $2m$ joining edges of F_2 are not hit, so the cost is $4m$ (the paper then lifts ρ to all blowups). Lean: `layoutEquiv` and `layout_adj_pathUnion` (Common), `lost_permMat` ($\mathrm{lost}(P_\rho) = 0$), and Step 3 gives $C_{\mathrm{ed}}(P_\rho) = 4m$, hence $d_{\mathrm{ed}}^{\mathrm{OT}} \leq 4m$ by `dOTEdit_le`.

Step 3: the edge count (p. 51, l. 1600–1602). The paper counts the number b of edges of F_1' that an alignment π of the k-blowups maps to non-edges and obtains $\|A^{\pi}_{F_1'} - A_{F_2'}\|_{(1)} = 4mk^2 + 4b$. Lean, for every $Q \in \mathrm{DS}$: with

$$\mathrm{lost}(Q) = \sum_{(v,w),(v',w')} Q(v,w)\, A_{F_1}(v,v')\,(1 - A_{F_2}(w,w'))\, Q(v',w') \;\geq 0$$

(`lost`, `lost_nonneg`), the identity $|a - b| = b - a + 2a(1-b)$ for $a, b \in \{0,1\}$ and the row and column sums of Q give

$$\|B(F_1, F_2, Q)\|_{(1)} = \textstyle\sum A_{F_2} - \sum A_{F_1} + 2\,\mathrm{lost}(Q) = 4m + 2\,\mathrm{lost}(Q)$$

(`entrywiseNorm_B_eq`, `sum_adj_sub_sum_adj`). This is the paper's identity with b/k^2 replaced by $\mathrm{lost}(Q)/2$ (lost counts ordered pairs).

Step 4: NO-direction, clouds go into one path (l. 1598–1616). The paper sets $\gamma = 1/(2mA)$, takes an alignment of cost at most $(4m + 4\gamma)k^2$, deduces $b \leq \gamma k^2$, so that each cloud has a unique target path $j(v)$ receiving $\geq (1-\gamma)k$ of its vertices, and that $j(\cdot)$ is constant along edges. Lean: since the minimum is attained (`exists_dOTEdit_eq`), an optimal Q has cost $\leq 4m$, hence $\mathrm{lost}(Q) = 0$ exactly; so $Q(v,w)Q(v',w') = 0$ whenever $vv' \in E(F_1)$ and $ww' \notin E(F_2)$ (`mul_eq_zero_of_lost_eq_zero`). As every v has a neighbour, the whole row of v lies in one path of F_2, the same for adjacent vertices, hence for all vertices of a path of F_1 (`eq_of_pathUnion_adj`). These steps are inside the proof of `dOTEdit_le_iff` (`hcomp`, `hc`, `hconst`).

Step 5: counting (l. 1617–1620). The paper shows $|A - S_j| < \gamma m A = \frac{1}{2}$, so $S_j = A$ and, as $A/4 < a_i < A/2$, exactly three paths go to path j. Lean: `isYes_of_isDS` sums the column sums of Q over path j of F_2 and gets $S_j = A$ exactly; `card_eq_three_of_sum_eq` gives the triples.

5 Assumptions and trusted base

No hypotheses: the result is unconditional and uses only the axioms `propext`, `Classical.choice`, `Quot.sound` (checked by `scripts/CheckAxioms.lean`). The complexity side (polynomial size of the construction, strong NP-hardness of Three-Partition [GJ79]) is not formalised. Proposition 57 (p. 57, l. 1805–1808: membership of $\textsc{GSim}(\delta_{\mathrm{ed}}^{\mathrm{OT}})$ in NP and exact finite blowups) is not formalised.

6 Notes on the formalisation

- **No blowups, nothing cited.** The paper argues with blowups and needs $d_{\mathrm{ed}}^{\mathrm{OT}} = d_{\mathrm{ed}}^{\bullet}$ from [Gro25, Theorem 3.10] (p. 50, l. 1585–1586). Theorem 1 of the paper, which is formalised (`GsimOT.theorem1`), covers only the operator norms, not the edit distance, so following the paper would introduce a cited hypothesis. The Lean proof is the fractional form of the same counting ($b/k^2 \leftrightarrow \mathrm{lost}(Q)/2$, the cloud fractions $n_j(v)/k \leftrightarrow$ the row of Q), and the γ-approximation becomes exact because the minimum over DS is attained.
- **Hypotheses.** `dOTEdit_le_iff` assumes $a_i \geq 2$ for all i; `dOTEdit_shift_le_iff` applies it to the shifted instance (Step 0) and holds for every instance.
- The Lean names `dOTEdit` and `costEdit` stand for the paper's $d_{\mathrm{ed}}^{\mathrm{OT}}$ and C_{ed}.

7 How to check this result

From the repository root (after `export PATH=$HOME/.elan/bin:$PATH`):

- build the module:
  `lake build GsimOT.Theorem16_EditHardness.Theorem16`
  (or, after a full build, `lake env lean GsimOT/Theorem16_EditHardness/Theorem16.lean`);
- check the axioms: `bash build.sh` runs `scripts/CheckAxioms.lean`, which contains
  `#print axioms GsimOT.Hardness.ThreePartition.dOTEdit_le_iff`
  `#print axioms GsimOT.Hardness.ThreePartition.dOTEdit_shift_le_iff`;
- the table `Theorem16_correspondence.pdf` in this folder lists every paper item with its Lean declaration and source line.